

Lecture 6: Immersions and parametrized surfaces

We now pass from curves to surfaces.

For curves, regularity meant that the derivative never vanished. For surfaces, the corresponding condition is that two independent tangent directions survive under the derivative.

Definition. Let $U \subset \mathbb{R}^2$ be open. A smooth map

$$f : U \longrightarrow \mathbb{R}^3$$

is an ***immersion*** if its derivative

$$Df_u : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$$

is injective for every $u \in U$.

Equivalently,

$$\text{rank } Df_u = 2.$$

An immersion may have self-intersections. The condition concerns the derivative, not the global behavior of the map.

Write the coordinates on U as (u, v) , and write

$$f(u, v) = (x(u, v), y(u, v), z(u, v)).$$

The columns of the derivative are

$$f_u = \frac{\partial f}{\partial u}, \quad f_v = \frac{\partial f}{\partial v}.$$

Thus f is an immersion precisely when

$$f_u(u, v), f_v(u, v)$$

are linearly independent at every point. In $\mathbb{R}^3$, this is equivalent to

$$f_u \times f_v \neq 0.$$

A point where this condition fails is called a ***singular point*** of the parametrization.

Example. Let $h : U \rightarrow \mathbb{R}$ be smooth. Its graph is parametrized by

$$f(u, v) = (u, v, h(u, v)).$$

Then

$$f_u = (1, 0, h_u), \quad f_v = (0, 1, h_v),$$

and

$$f_u \times f_v = (-h_u, -h_v, 1).$$

The last component is always 1, so the cross product never vanishes. Every graph is therefore regularly parametrized.

The graph example is special: projection onto the first two coordinates recovers the parameters. General parametrized surfaces need not be graphs globally.

Definition. A *parametrized surface element* in $\mathbb{R}^3$ is an immersion

$$f : U \longrightarrow \mathbb{R}^3, \quad U \subset \mathbb{R}^2 \text{ open.}$$

The map f is also called a parametrization.

The points $(u, v) \in U$ are the parameters, while $f(u, v)$ is the corresponding point in space.

Fixing one parameter gives the coordinate curves

$$u \longmapsto f(u, v_0), \quad v \longmapsto f(u_0, v).$$

Their velocity vectors at (u_0, v_0) are f_u and f_v .

Thus a coordinate grid in U is carried to a curvilinear grid on the surface element.

A parametrization contains more information than the geometric surface element: it includes a choice of coordinates.

Definition. Two parametrized surface elements

$$f : U \rightarrow \mathbb{R}^3, \quad \tilde{f} : \tilde{U} \rightarrow \mathbb{R}^3$$

are equivalent if there is a diffeomorphism

$$\phi : \tilde{U} \longrightarrow U$$

such that

$$\tilde{f} = f \circ \phi.$$

A ***non-parametrized surface element*** is an equivalence class of parametrized surface elements.

The map ϕ changes the coordinates, but not the geometric points described by the surface element.

Example. The paraboloid $z = x^2 + y^2$ can be described using Cartesian or polar coordinates. On its portion with $x > 0$, set

$$f(u, v) = (u, v, u^2 + v^2), \quad (u, v) \in U = (0, \infty) \times \mathbb{R},$$

$$\tilde{f}(r, \theta) = (r \cos \theta, r \sin \theta, r^2), \quad (r, \theta) \in V = (0, \infty) \times (-\pi/2, \pi/2).$$

The change of coordinates is

$$\phi(r, \theta) = (r \cos \theta, r \sin \theta), \quad \tilde{f} = f \circ \phi.$$

It is a diffeomorphism $V \rightarrow U$, with smooth inverse

$$\phi^{-1}(u, v) = \left(\sqrt{u^2 + v^2}, \arctan(v/u) \right).$$

Indeed,

$$D\phi = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad \det D\phi = r > 0.$$

Thus these are equivalent surface elements. The curves with fixed r are horizontal circular arcs. Those with fixed θ are arcs of vertical parabolas.

Why exclude $r = 0$? There $\tilde{f}_\theta = 0$, so the polar map is not an immersion. The full paraboloid is nevertheless smooth at the origin: its Cartesian graph parametrization remains regular.

Let $f : U \rightarrow \mathbb{R}^3$ be a parametrized surface element and let $u \in U$.

Definition. The *tangent space* of f at u is

$$T_u f = \text{im } Df_u \subset \mathbb{R}^3.$$

Since Df_u has rank 2, this is a two-dimensional vector subspace. In coordinates,

$$T_u f = \text{span}\{f_u(u), f_v(u)\}.$$

The corresponding affine tangent plane through $p = f(u)$ is

$$p + T_u f.$$

If the image is an embedded surface S near p , we write this space as $T_p S$.

Definition. A *tangent vector* to the surface element at u is a vector in $T_u f$.

If $\xi = (a, b) \in \mathbb{R}^2$, then

$$Df_u(\xi) = af_u(u) + bf_v(u).$$

Thus (a, b) are the coordinates of the tangent vector relative to the basis (f_u, f_v) .

Tangent vectors are also velocities of curves on the surface. If

$$\gamma : (-\varepsilon, \varepsilon) \rightarrow U, \quad \gamma(0) = u,$$

and $c = f \circ \gamma$, then

$$\dot{c}(0) = Df_u(\dot{\gamma}(0)) \in T_u f.$$

Conversely, every vector in $T_u f$ arises in this way.

Definition. The *normal space* at u is

$$N_u f = (T_u f)^\perp.$$

It is one-dimensional. Any nonzero vector in $N_u f$ is a normal vector.

The two unit normal vectors are

$$\nu(u) = \frac{f_u \times f_v}{\|f_u \times f_v\|} \quad \text{and} \quad -\nu(u).$$

Choosing one of these signs smoothly is called choosing an orientation of the surface element.

The tangent and normal spaces give an orthogonal decomposition

$$\mathbb{R}^3 = T_u f \oplus N_u f.$$

Suppose $\tilde{f} = f \circ \phi$, where

$$\phi(r, s) = (u(r, s), v(r, s)).$$

The chain rule gives

$$\begin{aligned}\tilde{f}_r &= f_u u_r + f_v v_r, \\ \tilde{f}_s &= f_u u_s + f_v v_s.\end{aligned}$$

Since $D\phi$ is invertible, the two pairs

$$(f_u, f_v) \quad \text{and} \quad (\tilde{f}_r, \tilde{f}_s)$$

span the same tangent space.

Moreover,

$$\tilde{f}_r \times \tilde{f}_s = \det(D\phi) (f_u \times f_v) \circ \phi.$$

Thus the normal line is unchanged. The preferred unit normal changes sign precisely when ϕ reverses orientation.

A parametrized surface element need not be injective. Its image may cross itself, and two parameter points mapping to the same point of $\mathbb{R}^3$ may represent different tangent planes.

Nevertheless, every immersion is locally an embedding.

Local form of an immersion. If $f : U \rightarrow \mathbb{R}^3$ is an immersion and $u_0 \in U$, then there is a neighborhood $V \subset U$ of u_0 such that

$$f|_V : V \longrightarrow \mathbb{R}^3$$

is an embedding onto its image.

Idea of the proof. Two components of f have an invertible 2×2 Jacobian at u_0 . The inverse function theorem makes those two components into local coordinates. In these coordinates, the remaining component is a graph.

Definition. A subset $S \subset \mathbb{R}^3$ is a smooth **surface**, or an embedded two-dimensional submanifold, if every point $p \in S$ has a neighborhood in which S is the image of an embedding

$$f : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3.$$

Equivalently, for each $p \in S$ there is an open neighborhood $W \subset \mathbb{R}^3$ and a smooth function $F : W \rightarrow \mathbb{R}$ such that

$$S \cap W = F^{-1}(0), \quad \nabla F \neq 0 \text{ on } F^{-1}(0).$$

Thus:

- a parametrized surface element is a local description;
- an immersion may have global self-intersections;
- a submanifold is locally embedded and is covered by compatible parametrizations.

For example, the sphere

$$S_R^2 = \{(x, y, z) : x^2 + y^2 + z^2 = R^2\}$$

is a submanifold. Set

$$F(x, y, z) = x^2 + y^2 + z^2 - R^2.$$

Then

$$\nabla F(x, y, z) = 2(x, y, z),$$

which never vanishes on S_R^2 .

At $p \in S_R^2$, a tangent vector X satisfies

$$dF_p(X) = 2\langle p, X \rangle = 0.$$

Therefore

$$T_p S_R^2 = p^\perp,$$

and the radial vector p is normal to the sphere.

This description agrees with the tangent space obtained from any local parametrization.